

Problem

A balanced Y-Y four-wire system has phase voltages

$$\begin{aligned} \mathbf{V}_{an} &= 120\angle 0^\circ, & \mathbf{V}_{bn} &= 120\angle -120^\circ \\ \mathbf{V}_{cn} &= 120\angle 120^\circ \text{ V} \end{aligned}$$

The load impedance per phase is $19 + j13 \, \Omega$, and the line impedance per phase is $1 + j2 \, \Omega$. Solve for the line currents and neutral current.

Step-by-step solution

Step 1 of 6

The phase voltages of balanced Y-Y four wire system are,

$$\begin{aligned} \mathbf{V}_{an} &= 120\angle 0^\circ \text{ V} \\ \mathbf{V}_{bn} &= 120\angle -120^\circ \text{ V} \\ \mathbf{V}_{cn} &= 120\angle 120^\circ \text{ V} \end{aligned}$$

The load impedance per phase, $\mathbf{Z}_L$ is $19 + j13 \, \Omega$

The line impedance per phase, $\mathbf{Z}_l$ is $1 + j2 \, \Omega$

Step 2 of 6

Calculate the value of total load impedance per phase, $\mathbf{Z}_Y$.

$$\mathbf{Z}_Y = \mathbf{Z}_L + \mathbf{Z}_l$$

Substitute $19 + j13 \, \Omega$ for $\mathbf{Z}_L$ and $1 + j2 \, \Omega$ for $\mathbf{Z}_l$.

$$\begin{aligned} \mathbf{Z}_Y &= 19 + j13 + 1 + j2 \\ &= 20 + j15 \, \Omega \end{aligned}$$

Step 3 of 6

Calculate the line currents of balanced Y-Y four wire system, $\mathbf{I}_a$.

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y}$$

Substitute $20 + j15$ for $\mathbf{Z}_Y$ and $120\angle 0^\circ$ for $\mathbf{V}_{an}$.

$$\begin{aligned} \mathbf{I}_a &= \frac{120\angle 0^\circ}{20 + j15} \\ &= \frac{120\angle 0^\circ}{25\angle 36.87^\circ} \\ &= 4.8\angle -36.87^\circ \text{ A} \end{aligned}$$

Step 4 of 6

Calculate the line currents of balanced Y-Y four wire system, $\mathbf{I}_b$.

$$\mathbf{I}_b = \frac{\mathbf{V}_{bn}}{\mathbf{Z}_Y}$$

Substitute $20 + j15$ for $\mathbf{Z}_Y$ and $120\angle -120^\circ$ for $\mathbf{V}_{bn}$.

$$\begin{aligned}\mathbf{I}_b &= \frac{120\angle -120^\circ}{20 + j15} \\ &= \frac{120\angle -120^\circ}{25\angle 36.87^\circ} \\ &= 4.8\angle -156.87^\circ \text{ A}\end{aligned}$$

Step 5 of 6

Calculate the line currents of balanced Y-Y four wire system, $\mathbf{I}_c$.

$$\mathbf{I}_c = \frac{\mathbf{V}_{cn}}{\mathbf{Z}_Y}$$

Substitute $20 + j15$ for $\mathbf{Z}_Y$ and $120\angle 120^\circ$ for $\mathbf{V}_{cn}$.

$$\begin{aligned}\mathbf{I}_c &= \frac{120\angle 120^\circ}{20 + j15} \\ &= \frac{120\angle 120^\circ}{25\angle 36.87^\circ} \\ &= 4.8\angle 83.13^\circ \text{ A}\end{aligned}$$

Therefore, the line currents in a balanced Y-Y four wire system are,

$$\begin{aligned}\mathbf{I}_a &= 4.8\angle -36.87^\circ \text{ A} \\ \mathbf{I}_b &= 4.8\angle -156.87^\circ \text{ A} \\ \mathbf{I}_c &= 4.8\angle 83.13^\circ \text{ A}\end{aligned}$$

Step 6 of 6

Calculate neutral current value in a balanced Y-Y four wire system.

$$\begin{aligned}\mathbf{I}_n &= -(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c) \\ &= -(4.8\angle -36.87^\circ + 4.8\angle -156.87^\circ + 4.8\angle 83.13^\circ) \\ &= -(3.84 - j2.88 - 4.41 - j1.885 + 0.57 + j4.765) \\ &= 0 \text{ A}\end{aligned}$$

Therefore, the neutral current in a balanced Y-Y four wire system is

$$\mathbf{I}_n = 0 \text{ A}.$$